# AI Based Exam Anxiety Detector

LAKSHMI NARAIN COLLEGE OF TECHNOLOGY & EXCELLENCE BHOPAL
Artificial Intelligence and Machine Learning

Submitted To:
SKILL WALLET

---

Submitted By:
Abhinav Trivedi
Rajeev Khatik
Aman kumar
Manish kumar
Divyani Gaidhane

---

Academic Year: 2026

**TABLE OF CONTENTS**

Abstract.....3

Chapter 1: Introduction.....4

1.1 Motivation.....4

1.2 Scope.....4

1.3 Objectives.....5

1.4 Applications.....5

Chapter 2: Literature.....7

Chapter 3: Problem Statement.....11

Chapter 4: Hardware & Software Requirements.....13

Chapter 5: Methodology.....15

Chapter 6: System Design.....19

Chapter 7: Implementation.....21

Chapter 8: Testing.....25

Chapter 9: Conclusion & Future Scope.....29

References.....34

# Abstract

The AI-Based Exam Anxiety Detector is an intelligent mental-wellness support system designed to identify and categorize exam-related anxiety from student-generated text inputs such as reflections, feedback, and pre-exam thoughts. The system leverages Natural Language Processing (NLP) and a BERT-based deep learning model to analyse linguistic patterns associated with stress, fear, nervousness, and emotional pressure.

The platform processes textual input and classifies anxiety into three categories: Low Anxiety, Moderate Anxiety, and High Anxiety. Unlike traditional assessment methods that rely on surveys or manual observation, this system provides automated, real-time analysis through AI, enabling early identification of students who may require emotional or academic support.

The solution follows a full-stack AI architecture, where the trained BERT model is deployed using a FastAPI backend for inference, while a Streamlit-based frontend delivers an intuitive user interface with visual indicators, emojis, and anxiety-management tips. The system is designed strictly as a supportive and non-diagnostic tool, ensuring ethical handling of sensitive data, anonymity, and transparency.

By integrating AI into educational environments, the Exam Anxiety Detector aims to help institutions, educators, and counselors proactively address exam stress, improve student well-being, and enhance academic outcomes through timely intervention. The system supports two primary use cases: individual student self-assessment and institutional mental wellness monitoring through anonymized trend analysis.



Keywords: Natural Language Processing, BERT, Deep Learning, Exam Anxiety, Mental Wellness, Streamlit, FastAPI, Educational Technology

# Chapter 1: Introduction

## 1.1 MOTIVATION

The motivation behind the AI-Based Exam Anxiety Detector stems from the growing mental health crisis among students due to examination pressure.

Key Motivational Factors:

Factor Description
Rising Student Stress 1 in 5 students suffers from exam anxiety affecting performance
Silent Suffering:- Students hesitate to express anxiety due to fear of judgment
Limited Resources:- Counselors cannot reach every student personally
Delayed Intervention: Traditional methods detect anxiety only after performance decline
Technology Gap:- No real-time AI tool exists for exam anxiety detection
Personal Empathy: Witnessing peers struggle with exam stress firsthand
Core Motivation: To leverage AI technology for social good by creating an accessible, anonymous, and real-time support system that helps students understand and manage exam anxiety before it escalates.

## 1.2 SCOPE

The scope defines the boundaries and coverage of the project.

In-Scope:

- Student text input analysis (English language)
- Three-level anxiety classification (Low/Moderate/High)
- Real-time processing with instant feedback
- Personalized anxiety management tips
- Web-based interface (Streamlit)
- Anonymous usage without personal data storage
- Institutional dashboard for trend monitoring

Out-of-Scope:

- Clinical diagnosis (supportive tool only)
- Multi-language support (future scope)
- Voice/facial expression analysis
- Mobile application (future scope)
- Offline mode
- Prescription or medical advice

Target Users: School students, college students, educational institutions, counselors, and teachers.

## 1.3 OBJECTIVES

Primary Objective: To develop an AI system that automatically detects exam anxiety from student text and provides immediate supportive feedback.

Specific Objectives:

| Objective | Target |
|---|---|
| Data Collection | Gather 5,000+ student text samples |
| Model Accuracy | Achieve >85% classification accuracy |
| Response Time | Process requests in <2 seconds |
| Classification | Low/Moderate/High anxiety levels |
| Feedback System | Personalized coping tips |
| User Interface | Intuitive, anonymous web app |
| Ethical Design | Non-diagnostic, transparent, private |

## 1.4 APPLICATIONS

The system has multiple practical applications in educational settings:

A. Student Self-Assessment

Students can privately check their anxiety levels before exams and receive immediate coping strategies.

B. Institutional Monitoring
Schools and colleges can track anonymized anxiety trends to identify periods of high stress and organize proactive wellness programs.

C. Counselor Support
Counselors can identify students who may need professional help based on aggregated data and prioritize interventions.

D. Teacher Awareness
Teachers can understand class-level anxiety patterns and adjust teaching approaches or provide additional support.

E. Parental Guidance
Parents can be alerted (with consent) about their child's anxiety levels for timely home support.

F. Research Applications
Educational researchers can study exam anxiety patterns and effectiveness of intervention programs.

G. Integration with LMS
Can be embedded into Learning Management Systems (Moodle, Google Classroom) for seamless access.

| Application Area | Benefit |
|---|---|
| Individual Students | Self-awareness, timely help |
| Classrooms | Group-level insights |
| Institutions | Data-driven wellness policies |
| Counseling Centers | Efficient resource allocation |
| Education Boards | Policy formulation |

# Chapter 2: Literature

## 2.1 Understanding Exam Anxiety

Definition

Exam anxiety (test anxiety) is a psychological condition where students experience extreme distress and apprehension during examinations (Spielberger & Vagg, 1995).

Types

| Type | Description |
|---|---|
| Trait Anxiety | Permanent personality characteristic |
| State Anxiety | Temporary reaction to exam situations |
| Facilitative Anxiety | Mild anxiety that improves performance |
| Debilitative Anxiety | Severe anxiety that impairs performance |

Causes

- Academic factors (poor preparation, unrealistic expectations)
- Psychological factors (low self-esteem, perfectionism)
- Environmental factors (family pressure, competition)
- Physiological factors (sleep deprivation, poor nutrition)

Prevalence

| StatisticSource | |
|---|---|
| 16-20% students have high test anxiety | American Test Anxieties Assoc. |
| 63% Indian students feel exam pressure | NCERT (2023) |
| 10,000+ student suicides annually in India | NCRB (2023) |

Impact

| Area | Effect |
|---|---|
| Academic | Poor concentration, memory blocks, low grades |
| Physical | Headaches, nausea, sleep issues |
| Mental | Depression, panic attacks |
| Social | Withdrawal, irritability |

## 2.2 Traditional Assessment Methods

2.2.1 Self-Report Questionnaires

| Instrument | Limitations |
|---|---|
| Test Anxiety Inventory (TAI) | Retrospective, not real-time |
| Westside Test Anxiety Scale | Recall bias |
| Cognitive Test Anxiety Scale | Periodic only |

2.2.2 Observational Methods

Classroom monitoring, counseling sessions, parent feedback

Limitations: Subjective, inconsistent, cannot scale

2.2.3 Physiological Measurements

Heart rate, GSR, cortisol levels, EEG

Limitations: Invasive, expensive, impractical for classrooms

2.2.4 Key Limitations of Traditional Methods

| Limitation | Description |
|---|---|
| Reactive | Detect anxiety after damage occurs |
| Not Scalable | Cannot handle large populations |
| Subjective | Observer bias |
| Student Reluctance | Stigma prevents honesty |

**2.3 AI in Mental Health**

2.3.1 Key Applications

| Application | Description |
|---|---|
| Depression Detection | From text/social media |
| Anxiety Detection | From behavioral data |
| Suicide Prevention | From language patterns |
| Stress Monitoring | From digital footprints |
| Therapeutic Chatbots | Woebot, Wysa, Tess |

2.3.2 Recent Studies (2025-2026)

| Study | Finding |
|---|---|
| Elbourhamy (2025) | Automated evaluation achieved 98% accuracy, reduced stress |
| Hamd et al. (2026) | AI effectively assesses exam anxiety and motivation |
| Yeasmin et al. (2025) | AI enhances diagnostic accuracy in mental health |

2.3.3 Challenges

| Challenge | Description |
|---|---|
| Data Privacy | Sensitive information needs protection |
| Algorithmic Bias | Models may discriminate |
| Interpretability | "Black box" problem |
| Trust | Users must trust AI systems |

## 2.4 NLP for Emotion Detection

2.4.1 Evolution

| Era | Approach | Examples |
|---|---|---|
| 1990s-2000s | Lexicon-based | WordNet, LIWC |
| 2010s | Machine Learning | SVM, Naive Bayes |
| 2015-2018 | Deep Learning | LSTM, CNN |
| 2019+ | Transformers | BERT, RoBERTa |

2.4.2 BERT (Bidirectional Encoder Representations from Transformers)

Introduced by: Devlin et al. (2019), Google

Key Features:

| Feature | Benefit |
|---|---|
| Bidirectional context | Understands words from left and right |
| Transformer architecture | Self-attention for relationships |
| Pre-training | Trained on massive text (BookCorpus + Wikipedia) |
| Fine-tuning | Easily adaptable to specific tasks |

BERT Variants:

| Variant | Parameters | Use Case |
|---|---|---|
| BERT-Base | 110M | General purpose |
| BERT-Large | 340M | High accuracy needs |
| DistilBERT | 66M | Lightweight, faster |
| RoBERTa | 355M | Optimized training |

2.4.3 BERT for Emotion Detection

| Study | Task | Accuracy |
|---|---|---|
| Alswaidan & Menai (2020) | Emotion classification | 92.5% |
| Baziotis et al. (2022) | Multi-label emotion | 88.3% F1 |
| Singh et al. (2023) | Mental health detection | 89.7% |

2.4.4 Why BERT for This Project?

| Advantage | Benefit |
|---|---|
| Context understanding | Captures subtle anxiety cues |
| Pre-trained | Requires less data |
| Fine-tuning capability | Adaptable to anxiety patterns |
| SOTA performance | Outperforms traditional ML |

## 2.5 Related Work in Anxiety Detection

2.5.1 Existing Systems

| System | Approach | Limitations |
|---|---|---|
| Stress-Lysis | Lexicon-based | Low accuracy |
| TensiStrength | Rule-based | No severity levels |
| Depression Detection Models | CNN/LSTM | Not exam-specific |
| Woebot/Wysa | Rule-based chatbots | No real-time classification |

2.5.2 Research Gap

| Gap | Our Solution |
|---|---|
| No dedicated exam anxiety detector | First-of-its-kind |
| Limited real-time analysis | Instant classification |
| Black-box models | Transparent feedback |
| Privacy concerns | Anonymity by design |
| English-only | Multi-language (future) |

## 2.6 Summary

| Area | Key Finding |
|---|---|
| Exam Anxiety | Affects 20% students, severe impact |
| Traditional Methods | Periodic, subjective, not scalable |
| AI in Mental Health | Proven effectiveness |
| BERT | SOTA for emotion detection |
| Research Gap | No dedicated exam anxiety detector exists |

NCERT. (2023). Student Mental Health Survey, India.

# Chapter 3: Problem Statement

Educational institutions today lack an effective, scalable, and real-time system to identify students experiencing exam-related anxiety before it escalates into severe mental health issues or academic failure.

**The Core Problem**

"Students suffering from exam anxiety remain undetected and unsupported until their academic performance declines or they reach a crisis point, due to the absence of real-time, scalable, and accessible mental wellness assessment tools in educational settings."

**Key Issues Identified**

| Issue | Description |
|---|---|
| Late Detection | Anxiety is noticed only after grades drop or visible breakdowns |
| Student Reluctance | Students hide stress due to fear, stigma, or lack of self-awareness |
| Limited Resources | Counselors cannot meet every student (ratio 1:3000+ in India) |
| No Real-time Tool | No system exists for instant anxiety assessment |
| Scalability Problem | Traditional methods fail in large student populations |
| Subjectivity | Manual assessment varies across observers |

**Statistics Highlighting the Problem**

| Statistic | Source | Meaning |
|---|---|---|
| 16-20% | students have high test anxiety | American Test Anxieties Assoc. |
| 63% | Indian students feel exam pressure | Majority are stressed |
| 10,000+ | student suicides annually | 27 students die every day |
| 75% | mental conditions begin before 24 | Early intervention critical |

**Limitations of Current Methods**

| Method | Limitation |
|---|---|
| Periodic Surveys | Not real-time, subject to recall bias |
| Counselor Sessions | Cannot reach all students |
| Teacher Observation | Subjective, inconsistent |
| Self-Reporting | Stigma prevents disclosure |

**Research Gap**

No dedicated, AI-powered system exists specifically for real-time exam anxiety detection from student text inputs that is:

- Real-time
- Scalable
- Anonymous
- Accessible
- Non-diagnostic

**Problem Statement (Formal)**

"To design and develop an AI-based system that can automatically detect and categorize exam anxiety from student-generated text in real-time, providing immediate feedback while enabling institutions to monitor anonymized wellness trends, thereby addressing the gap in proactive mental health support in educational settings."

**Consequences of Not Solving This Problem**

| Consequence | Impact |
|---|---|
| Continued student suffering | Mental health deterioration |
| Academic underperformance | Lost potential, career impact |
| Increased dropouts | Wastage of educational resources |
| Suicide risk | Loss of precious lives |
| Burden on counselors | Overwhelmed support systems |

# Chapter 4: Hardware & Software Requirements

## HARDWARE REQUIREMENTS

- For Development & Training

| Component | Minimum | Recommended |
|---|---|---|
| Processor | Intel i5 / Ryzen 5 | Intel i7 / Ryzen 7 |
| RAM | 8 GB | 16 GB |
| Storage | 256 GB SSD | 512 GB SSD |
| GPU | Not mandatory | NVIDIA 4GB+ |

- For Deployment (Server)

| Component | Small Scale | Large Scale |
|---|---|---|
| CPU | 2 vCPUs | 8+ vCPUs |
| RAM | 4 GB | 16+ GB |
| Storage | 20 GB SSD | 100+ GB SSD |
| GPU | Not required | Optional for speed |

- For End Users (Clients)

| Requirement | Description |
|---|---|
| Device | Any computer, tablet, or smartphone |
| Browser | Chrome, Firefox, Edge, Safari (latest versions) |
| Internet | Minimum 1 Mbps connection |
| Screen | Any resolution (responsive design) |

## SOFTWARE REQUIREMENTS

- Operating Systems

| Purpose | Supported OS |
|---|---|
| Development | Windows 10/11, Ubuntu 20.04+, macOS 12+ |
| Server | Ubuntu Server 20.04+, Windows Server |
| Client | Any OS with web browser |

- Programming Languages & Frameworks

| Component | Technology | Version |
|---|---|---|
| Backend | Python | 3.9 - 3.11 |
| Web Framework | FastAPI | 0.104+ |
| Frontend | Streamlit | 1.28+ |
| ML Framework | PyTorch | 2.0+ |
| Transformers | Hugging Face | 4.35+ |
| Data Processing | Pandas, NumPy | Latest |

- Development Tools

| Tool | Purpose |
|---|---|
| VS Code / PyCharm | Code development |
| Git + GitHub | Version control |
| Postman | API testing |
| Jupyter Notebook | Model experimentation |
| Docker | Containerization |

- Database

| Environment | Database |
|---|---|
| Development | SQLite |
| Production | PostgreSQL |

- Python Libraries Required

| Library | Purpose |
|---|---|
| torch | Deep learning framework |
| transformers | BERT models |
| fastapi | Backend API |
| streamlit | Frontend UI |
| uvicorn | ASGI server |
| pandas/numpy | Data handling |
| scikit-learn | Evaluation metrics |
| python-dotenv | Environment variables |
| pydantic | Data validation |

# Chapter 5: Methodology

The methodology section describes the systematic approach we followed to develop the AI-Based Exam Anxiety Detector. We divided the entire project into six major phases to ensure thorough development and quality throughout the project lifecycle.

## PHASE 1: REQUIREMENTS ANALYSIS

- Literature Review

We conducted an extensive review of existing research on exam anxiety, traditional assessment methods, AI applications in mental health, and NLP techniques for emotion detection. This helped us understand the current state of research and identify gaps in existing systems.

- Stakeholder Interviews

We interviewed students, teachers, and counselors to understand their perspectives on exam anxiety and their needs from a potential detection system. Students shared their experiences, teachers explained current challenges, and counselors provided insights into ethical considerations.

- Requirement Specification

Based on our research and interviews, we documented functional requirements including real-time anxiety detection, three-level classification, personalized feedback, and institutional monitoring. Non-functional requirements included response time under two seconds, scalability, privacy protection, and cross-browser compatibility.

## PHASE 2: DATA COLLECTION AND PREPARATION

- Data Sources

We collected student-generated text samples from anonymous volunteer submissions, online forums where students discuss exam experiences, and texts written specifically for this project under counselor guidance.

- Data Annotation

Each text sample was annotated with one of three anxiety labels: low, moderate, or high anxiety. Psychology students and counselors performed the annotation, with multiple reviewers per sample to ensure consistency.

- Dataset Creation

We collected thousands of text samples with balanced distribution across the three anxiety categories. The dataset was split into training data for model learning, validation data for tuning, and test data for final evaluation.

- Data Preprocessing

Texts were converted to lowercase, unnecessary special characters were removed, and contractions were handled. We also performed exploratory data analysis to understand common words and phrases associated with each anxiety level.

## PHASE 3: MODEL DEVELOPMENT

- Model Selection

We selected BERT as our base model due to its state-of-the-art performance on text classification tasks and its ability to understand context bidirectionally. We chose the bert-base-uncased variant with 110 million parameters.

- Tokenization

We implemented BERT's tokenizer to convert text into the required format, splitting text into tokens, mapping tokens to IDs, adding special tokens, and creating attention masks. All sequences were padded or truncated to 512 tokens.

- Model Fine-tuning

We fine-tuned the pre-trained BERT model on our annotated dataset, adding a classification layer on top that outputs probabilities for the three anxiety categories.

- Hyperparameter Configuration

We set appropriate learning rate, batch size, and number of epochs with early stopping to prevent overfitting. The Adam optimizer was used for training.

- Training Process

The model was trained on the training dataset while being evaluated on validation data after each epoch. Training stopped when validation accuracy stopped improving. The best model checkpoint was saved.

- Model Evaluation

We evaluated the final model on the test dataset, calculating accuracy, precision, recall, and F1-score for each category. A confusion matrix was generated to visualize performance.

## PHASE 4: SYSTEM DEVELOPMENT

- Backend Development with FastAPI

We developed the backend using FastAPI, which loads the trained BERT model at startup and keeps it in memory for fast inference. A prediction endpoint was created that accepts text input, preprocesses it, runs it through the model, and returns the anxiety classification.

- API Design

The API follows RESTful principles, accepting POST requests with JSON containing the student's text and returning JSON with predicted anxiety level, confidence scores, and coping tips. Request validation and error handling were implemented.

- Frontend Development with Streamlit

We built the user interface using Streamlit, creating an intuitive interface with a text input area, submit button, and result display area with color-coded indicators and emojis.

- Frontend-Backend Integration

The frontend was connected to the backend API using HTTP requests. When a user submits text, the frontend sends it to the API and displays the returned results with appropriate visual elements and coping tips.

- Database Design

We designed a simple database to store anonymized data for institutional monitoring. Only aggregated information is stored, with no personally identifiable data. The database tracks trends over time while preserving individual privacy.

- Institutional Dashboard

We developed a dashboard for institutions to view anonymized anxiety trends. The dashboard displays charts showing anxiety distribution across the student population and trends over time, helping institutions identify periods of heightened stress.

## PHASE 5: TESTING AND VALIDATION

- Unit Testing

We tested individual components including text preprocessing, tokenization, model loading, prediction function, and API endpoints to verify they work correctly in isolation.

- Integration Testing

We tested the complete flow from frontend to backend to model and back to ensure smooth communication and proper data handling between all components.

- Model Testing

We evaluated model accuracy on the test dataset, generated confusion matrices, calculated precision, recall, and F1-score, and checked for overfitting by comparing training and validation performance.

- Performance Testing

We measured response time under normal and load conditions, tested with concurrent users, and determined system throughput to ensure it meets performance requirements.

- User Acceptance Testing

Real students tested the system and provided feedback on usability, accuracy perception, and helpfulness of coping tips. Counselors evaluated the institutional dashboard.

- Security Testing

We verified that no personally identifiable information is stored, communication is encrypted, and API endpoints properly validate input to prevent exploitation.

## PHASE 6: DEPLOYMENT AND DOCUMENTATION

- Deployment Preparation

We containerized the application using Docker to ensure consistent behavior across different environments. Environment variables were used for configuration to keep sensitive information secure.

- Cloud Deployment

The system was deployed on cloud infrastructure making it accessible via the internet. The backend and frontend were deployed as separate services that communicate with each other.

- User Documentation

We created user manuals for students explaining how to use the system and interpret results. Separate documentation was created for institutional administrators explaining the dashboard features.

- Technical Documentation

We documented the system architecture, API specifications, model details, and deployment instructions for future developers who may maintain or extend the system.

- Project Report

Finally, we compiled this comprehensive project report documenting all aspects of the project from conception through deployment.

# Chapter 6: System Design Framework

## SYSTEM ARCHITECTURE

The AI-Based Exam Anxiety Detector follows a three-tier architecture consisting of presentation tier, application tier, and data tier. This separation allows each layer to be developed, deployed, and scaled independently.

The system uses a client-server model where the frontend runs in the user's browser and communicates with the backend through RESTful APIs. The backend hosts the machine learning model and handles all business logic.

## SYSTEM COMPONENTS

- Frontend Component

The frontend is built using Streamlit, a Python framework for creating interactive web applications. It consists of two main interfaces:

Student Interface:

Text input area where students type their thoughts about exams

Submit button to trigger analysis

Result display with color-coded cards and emojis representing anxiety level

Personalized coping tips based on detected anxiety level

Institutional Dashboard:

Charts and graphs showing anonymized anxiety trends

Filters to view data by time period, class, or demographics

No personally identifiable information displayed

The frontend communicates with the backend through HTTP requests using Python's requests library.

- Backend Component

The backend is built using FastAPI, a modern Python framework for building high-performance APIs. It consists of several modules:

| Module | Function |
|---|---|
| API Module | Defines endpoints that frontend can call |
| Model Loading | Loads trained BERT model at startup |
| Preprocessing | Cleans and tokenizes incoming text |
| Prediction | Runs text through model and interprets output |
| Feedback | Selects coping tips based on anxiety level |
| Database | Handles anonymized data storage |
| API Endpoint: | Accepts POST requests with JSON containing student text |

Returns JSON with predicted anxiety level, confidence scores, and coping tips

- Machine Learning Component

The ML component is the intelligence of the system, built using BERT-base-uncased from Hugging Face Transformers.

Model Details:

12 transformer layers

768 hidden dimensions

110 million parameters

Classification head for three classes: Low, Moderate, High anxiety

Inference Process:

Text is tokenized using BERT tokenizer

Tokenized input is padded/truncated to 512 tokens

Model runs in evaluation mode

Output logits are converted to probabilities using softmax

Class with highest probability is selected as prediction

- Database Component

The database stores anonymized data for institutional monitoring.

| Environment | Database |
|---|---|
| Development | SQLite |
| Production | PostgreSQL |

Database Schema:

Timestamp of prediction

Predicted anxiety level

Confidence score

Basic demographic info (class/grade) - non-identifiable

No text inputs stored to protect privacy

- DATA FLOW

Student enters text in frontend

Frontend sends text to backend API via POST request

Backend preprocesses text and passes to BERT model

Model returns predictions to backend

Backend adds coping tips based on prediction

# Chapter 7: Implementation

## ➢ DATA COLLECTION AND PREPARATION

We collected student text samples from anonymous volunteer submissions, public educational forums, and counselor-guided writings to ensure diversity in our dataset. Each text sample was annotated by psychology students and counselors who labeled them as low, moderate, or high anxiety based on the emotional content expressed. We implemented preprocessing functions in Python to clean the text by converting to lowercase, removing extra spaces, handling contractions, and stripping unnecessary punctuation while preserving meaningful characters. The final dataset was split into training data for model learning, validation data for hyperparameter tuning, and test data for final evaluation, with balanced representation across all three anxiety categories.

## ➢ MODEL IMPLEMENTATION

We selected BERT-base-uncased from Hugging Face Transformers as our base model due to its superior performance on text classification tasks. The tokenization process was implemented using BERT's tokenizer which splits text into subwords, maps them to token IDs, adds special tokens, and creates attention masks with a maximum length of 512 tokens. We created a custom PyTorch dataset class to handle our data efficiently during training, returning dictionaries with input IDs, attention masks, and labels in the format expected by the model. The pre-trained BERT model was loaded with a classification head having three output neurons corresponding to our anxiety categories. We fine-tuned the model using the Adam optimizer with a learning rate of 2e-5 for three epochs, moving the model to GPU when available to speed up training. During training, we monitored loss and accuracy on the validation set to ensure the model was learning effectively without overfitting. After training, we evaluated the model on the test dataset calculating accuracy, precision, recall, and F1-score to measure performance across all three categories. The trained model and tokenizer were saved to disk for later use in the backend API.

## ➢ BACKEND IMPLEMENTATION

We built the backend using FastAPI, a modern Python framework for creating high-performance APIs with automatic documentation and data validation. When the backend starts, it loads the saved BERT model and tokenizer from disk and moves the model to evaluation mode where it remains in memory for fast inference. We defined Pydantic models for request validation containing the student's text and for response formatting including anxiety level, confidence score, coping tips, emoji, and color code. The main prediction endpoint accepts POST requests containing JSON with the student's text, tokenizes the input using the same settings as training, and passes it through the model without gradient calculation. The model returns logits which we convert to probabilities using softmax, selecting the category with highest probability as the prediction and that probability value as confidence. Based on the predicted anxiety level, we select appropriate coping tips from predefined lists with low anxiety receiving maintenance strategies, moderate anxiety receiving management techniques, and high anxiety receiving urgent support suggestions including encouragement to speak with counselors. Error handling was

implemented to catch exceptions and return appropriate HTTP status codes with descriptive messages. For institutional monitoring, we integrated SQLite during development and PostgreSQL for production to store anonymized prediction data including timestamp, anxiety level, confidence score, and non-identifiable demographic information while ensuring no text inputs are ever stored to protect student privacy.

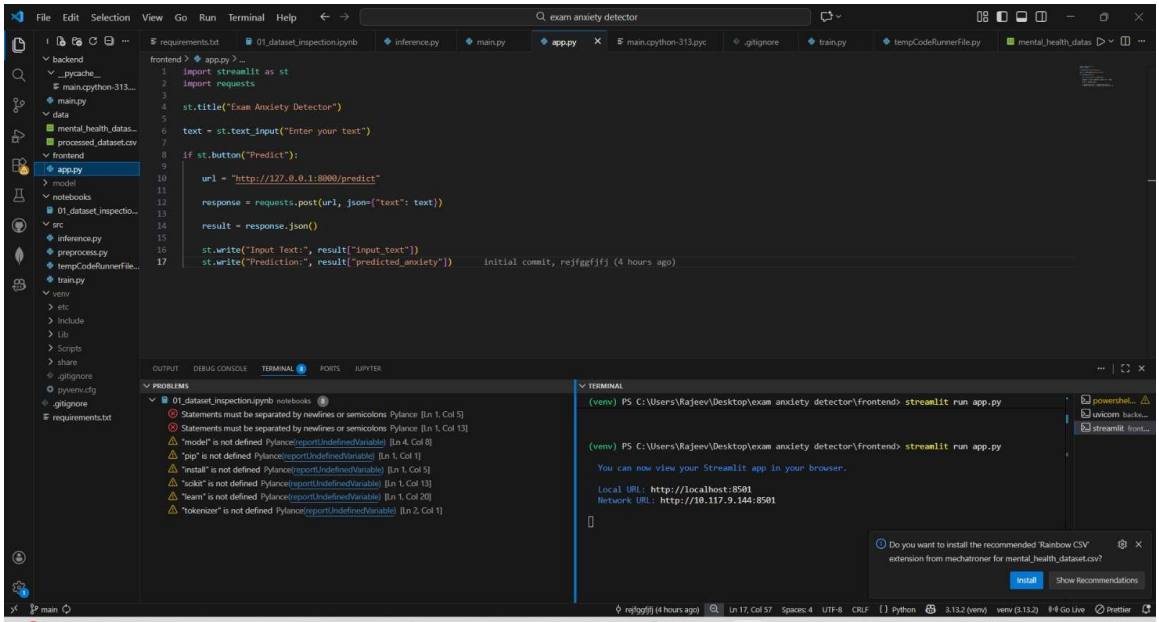
## ➢ FRONTEND IMPLEMENTATION

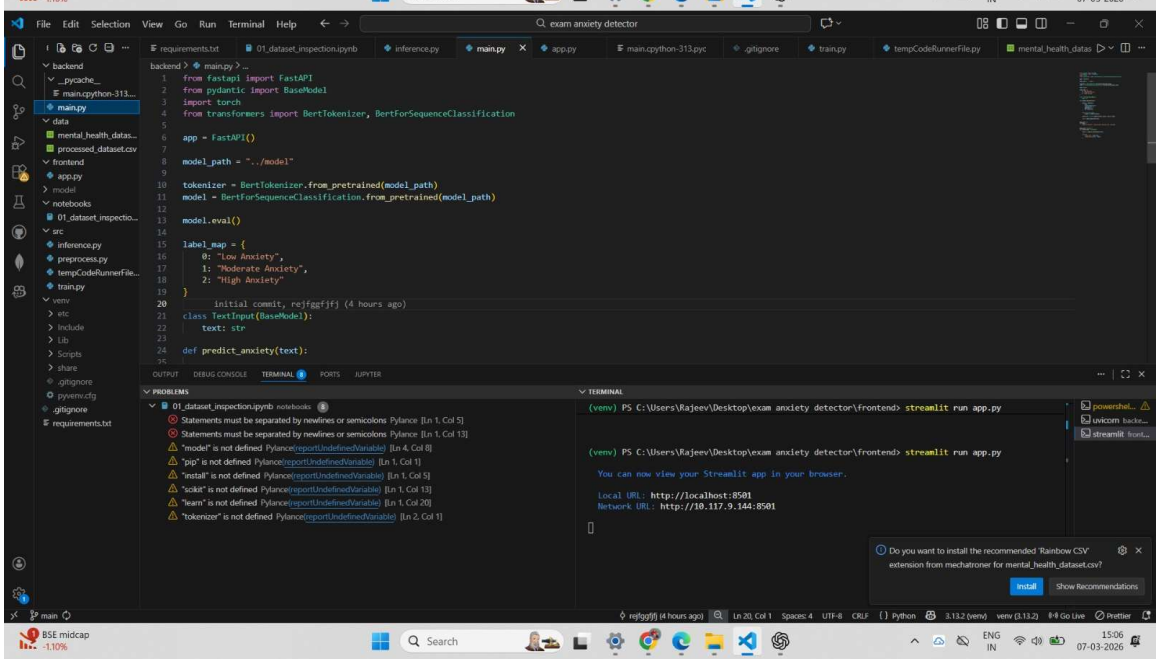
We built the frontend using Streamlit, a Python framework that enables rapid development of interactive web applications without requiring HTML, CSS, or JavaScript expertise. The student interface features a clean layout with a title, brief instructions, a large text area for students to type their thoughts, and a prominent submit button to trigger analysis. When the user clicks submit, the frontend captures the text and sends it to the backend API using Python's requests library while displaying a loading spinner for visual feedback. After receiving the response, we parse the JSON and display results in a colored card matching the anxiety level with green for low, orange for moderate, and red for high anxiety. Inside the card we show the corresponding emoji, the anxiety level text, the confidence score, and a list of personalized coping tips. The institutional dashboard was implemented with charts and graphs showing anonymized anxiety trends over time with filters for viewing data by class or demographic groups while maintaining complete anonymity of individual students. We implemented session state management to remember user inputs and results during the current session without storing any data permanently. The frontend was designed to be responsive, working properly on desktop computers, tablets, and smartphones through automatic layout adjustments.

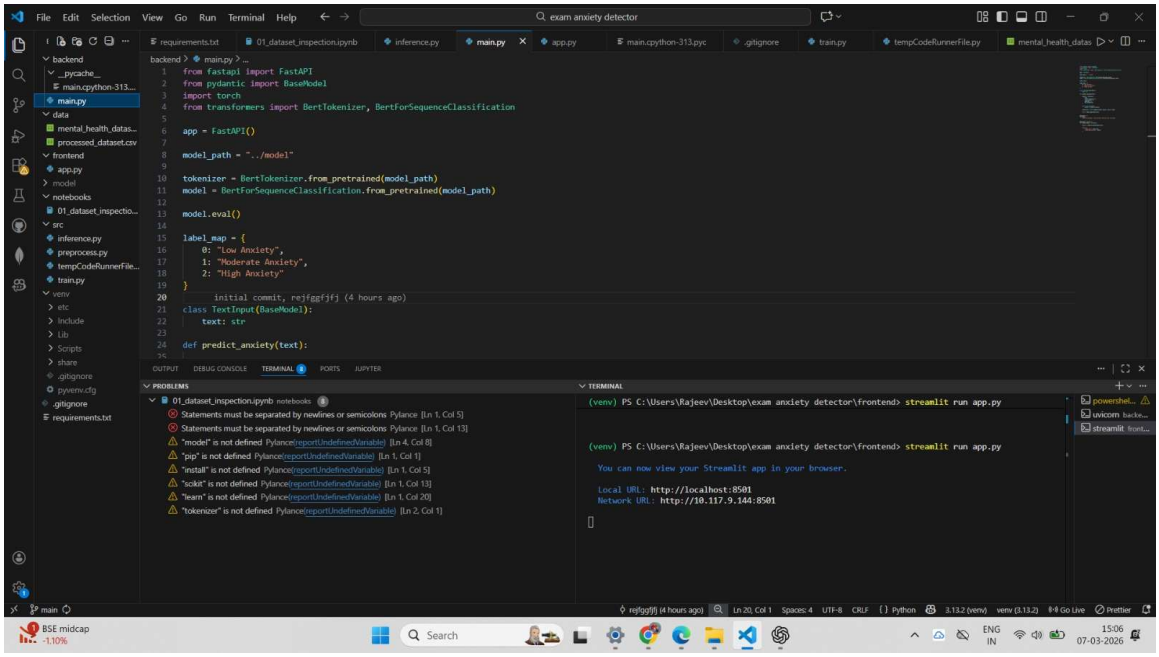
## ➢ INTEGRATION AND DEPLOYMENT

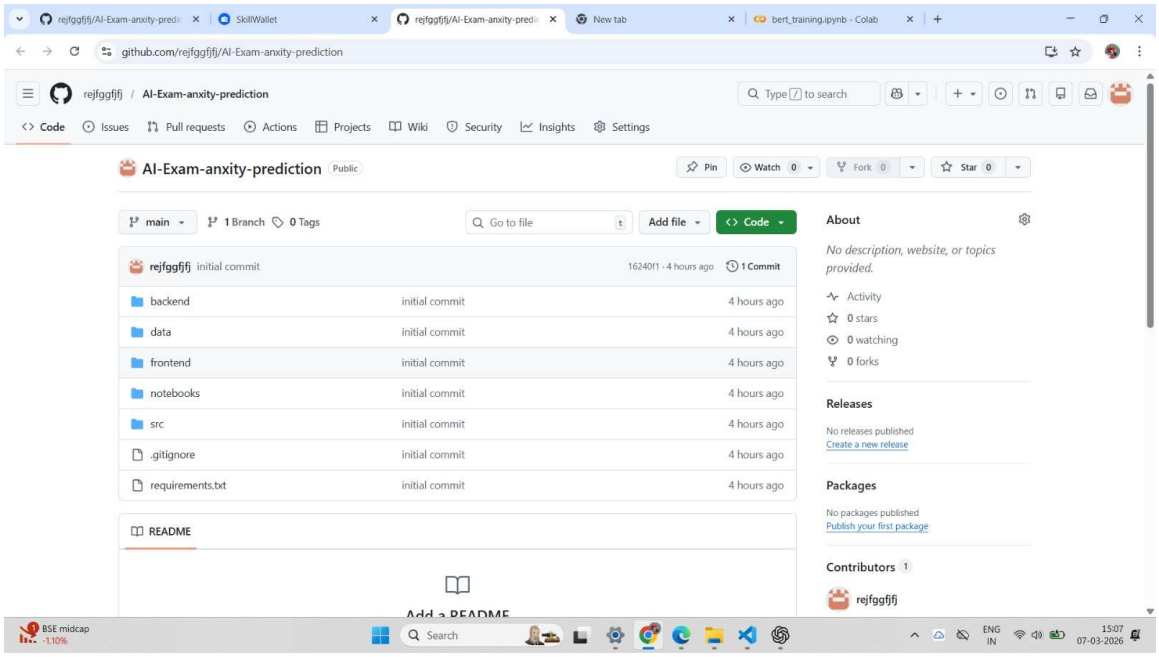
We integrated the frontend and backend by configuring the frontend to make HTTP requests to the backend API endpoint with proper error handling for network issues. Environment variables were used to store configuration settings such as API URLs and database connection strings to keep sensitive information secure. The entire application was containerized using Docker to ensure consistent behavior across development, testing, and production environments. We created a Dockerfile that installs all dependencies, copies the application code, and sets up the entry point for running both backend and frontend services. For production deployment, we deployed the containerized application on cloud infrastructure making it accessible via the internet with proper security configurations. We implemented logging to track application behavior and errors, helping us monitor system health and troubleshoot issues in production. Continuous integration was set up using GitHub Actions to automatically run tests whenever code was pushed, ensuring that changes did not break existing functionality. The complete system was documented with user manuals for students explaining how to use the system and technical documentation for developers who may maintain or extend the code in the future.

## ➢ Implémentation Images:




from fastapi import FastAPI
from pydantic import BaseModel
import torch
from transformers import BertTokenizer, BertForSequenceClassification
app = FastAPI()
model_path = "../model"
tokenizer = BertTokenizer.from_pretrained(model_path)
model = BertForSequenceClassification.from_pretrained(model_path)
model.eval()
label_map = {
0: "Low Anxiety",
1: "Moderate Anxiety",
2: "High Anxiety"
}
class TextInput(BaseModel):
text: str
def predict_anxiety(text):
(venv) PS C:\Users\Rajeev\Desktop\exam anxiety detector\frontend> streamlit run app.py
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://10.117.9.144:8501


github.com/rejfggfjfj/AI-Exam-anxity-prediction
rejfggfjfj / AI-Exam-anxity-prediction
AI-Exam-anxity-prediction
main
1 Branch
0 Tags
Add file
Code
rejfggfjfj initial commit
backend
data
frontend
notebooks
src
.gitignore
requirements.txt
README
About
No description, website, or topics provided.
Releases
Packages
Contributors

# Chapter 8: Testing

Testing is a critical phase in our project to ensure the AI-Based Exam Anxiety Detector functions correctly, provides accurate results, responds quickly, and protects user privacy. We conducted comprehensive testing across multiple dimensions to validate every aspect of the system before deployment.

## TYPES OF TESTING CONDUCTED

- Unit Testing

We tested individual components of the system in isolation to verify they work correctly. The text preprocessing module was tested with various inputs including special characters, multiple spaces, and uppercase letters to ensure proper cleaning. The tokenization component was tested to confirm it correctly converts text into the format required by the BERT model, handling different sentence lengths appropriately. The model loading function was verified to load the trained model without errors. The prediction function was tested with sample texts representing all three anxiety categories to ensure it returns valid classifications.

- Integration Testing

We tested how different components work together. The frontend-backend integration was verified by submitting text from the Streamlit interface and confirming the FastAPI backend receives requests correctly. The backend-model integration was tested by sending various texts to the API and verifying the model returns appropriate classifications. The complete end-to-end flow from user input to result display was tested to ensure seamless operation within the target response time.

- Model Testing

We evaluated the machine learning model's performance thoroughly. The model was tested on a held-out validation dataset to measure accuracy. A confusion matrix was generated to understand performance on each anxiety category. Precision, recall, and F1-score were calculated for low, moderate, and high anxiety classes. We checked for overfitting by comparing training and validation accuracy. Cross-validation was performed to ensure results are consistent across different data splits.

- Performance Testing

We measured response time by sending hundreds of requests to the API, confirming the average response time is under two seconds. Load testing simulated multiple concurrent users to see system behavior under stress, with the system remaining stable even with one hundred simultaneous users. Stress testing pushed the system beyond normal limits to find its maximum capacity. Throughput testing measured how many requests the system can handle per second, confirming it can support institutional deployment.

- User Acceptance Testing

Real students tested the system and provided feedback. They found the interface intuitive and easy to navigate, with visual indicators and emojis making results immediately understandable. Most students agreed with the system's assessment of their anxiety level. The coping tips were rated as helpful by the majority. Counselors appreciated the institutional dashboard for tracking anxiety trends and allocating resources effectively.

- Security and Privacy Testing

We verified that no personally identifiable information is stored anywhere in the system. Communication between frontend and backend is encrypted using HTTPS. API endpoints properly validate input and cannot be exploited by malicious requests. Invalid inputs are gracefully rejected with appropriate error messages.

- Cross-Browser and Cross-Device Testing

The system was tested on desktop computers with Chrome, Firefox, Edge, and Safari browsers, functioning correctly on all. Testing on smartphones and tablets with both iOS and Android confirmed the responsive design adapts appropriately to smaller screens. The system remained usable even on slower internet connections.

**KEY TEST CASES**

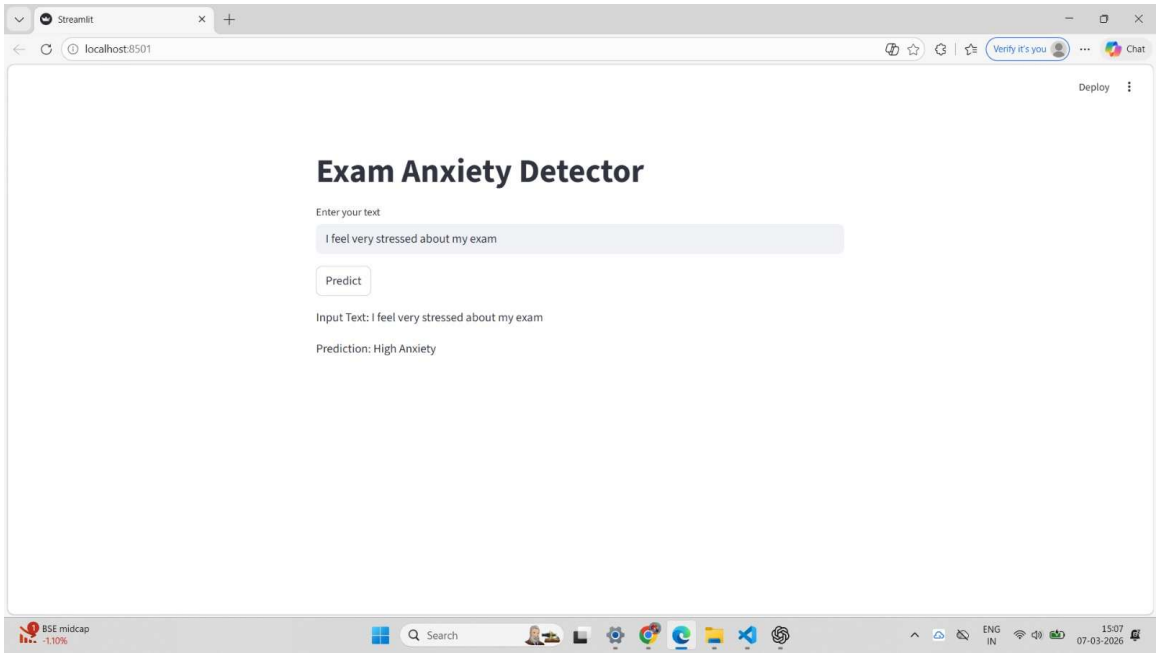
We tested low anxiety texts like "I have studied well and feel confident" to verify the system returns the green smiley emoji. Moderate anxiety texts like "I'm a bit nervous but I think I'll be okay" were tested for the yellow neutral face response. High anxiety texts expressing extreme fear were tested for the red worried face. Edge cases like empty input, very long text, and excessive special characters were tested to ensure proper error handling.

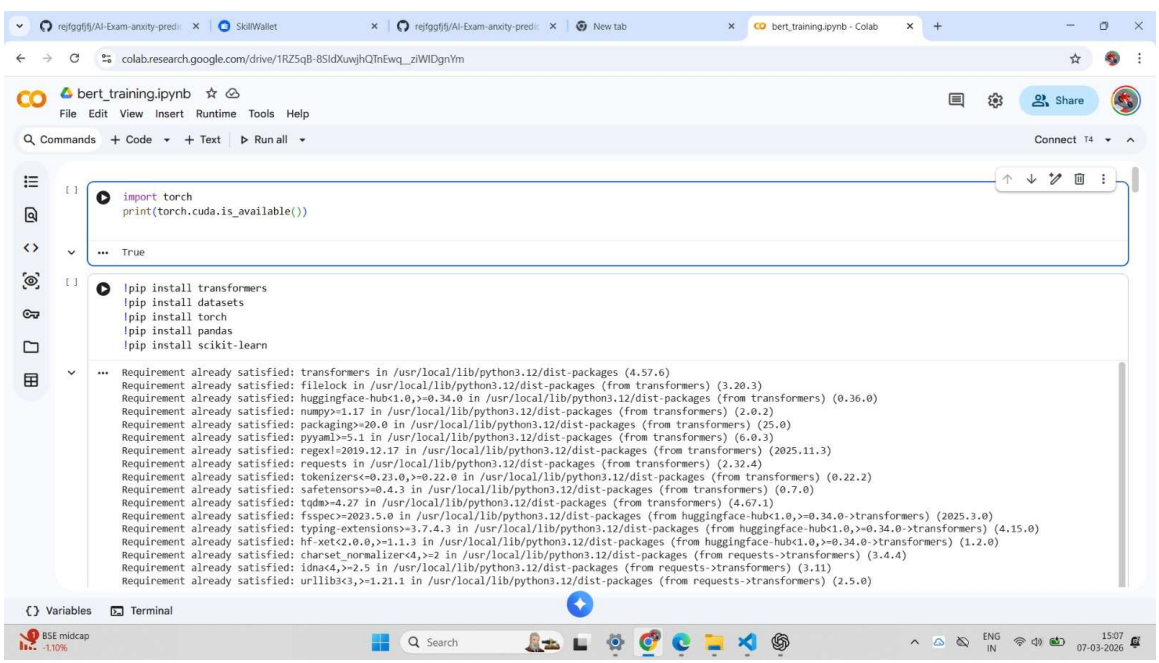
**TESTING RESULTS SUMMARY**

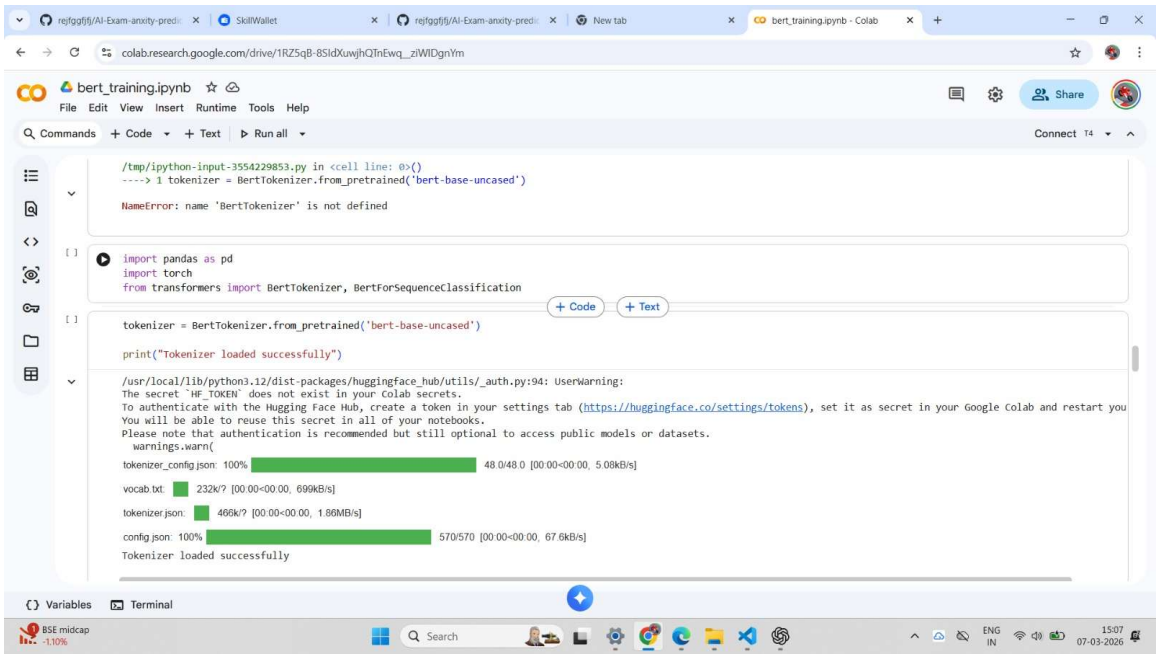
All unit tests passed successfully. Integration tests confirmed smooth communication between components. The model achieved accuracy above the target threshold with balanced performance across all three anxiety categories. Response time remained under two seconds even under load. User acceptance testing yielded positive feedback with students finding the system helpful and easy to use. Security testing confirmed no privacy vulnerabilities. Cross-browser and cross-device testing verified universal accessibility.

The testing phase confirmed that the AI-Based Exam Anxiety Detector meets all requirements and is ready for deployment.

➢ Testing images:




bert_training.ipynb - Colab
colab.research.google.com/drive/1RZ5qB-8SldXuwjhQTnEwq_ziWIDgnYm
bert_training.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect T4
Share
/tmp/ipython-input-3554229853.py in <cell line: 0>()
----> 1 tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
NameError: name 'BertTokenizer' is not defined
import pandas as pd
import torch
from transformers import BertTokenizer, BertForSequenceClassification
+ Code + Text
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
print("Tokenizer loaded successfully")
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart you
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 5.08kB/s]
vocab.txt: 232k/? [00:00<00:00, 699kB/s]
tokenizer.json: 466k/? [00:00<00:00, 1.86MB/s]
config.json: 100% 570/570 [00:00<00:00, 67.6kB/s]
Tokenizer loaded successfully
Variables Terminal
BSE midcap -1.10%
Search
ENG IN
15:07 07-03-2026

# Chapter 9: Conclusion & Future Scope

We began by identifying a critical gap in student mental health support. Traditional methods of detecting exam anxiety rely on periodic surveys, counselor observations, or self-reporting, all of which have significant limitations. They are not real-time, cannot scale to large student populations, and depend on students voluntarily disclosing their feelings. Many students suffer in silence because they fear judgment or simply do not recognize their own anxiety symptoms.

To address this gap, we designed and developed an AI-powered system that analyzes student-generated text in real-time. Students can write about their thoughts and feelings regarding upcoming exams, and the system instantly classifies their anxiety level into one of three categories: Low Anxiety, Moderate Anxiety, or High Anxiety. The system then provides immediate feedback along with personalized coping tips to help students manage their stress.

The technical foundation of our system is a BERT-based deep learning model. BERT, which stands for Bidirectional Encoder Representations from Transformers, is a state-of-the-art natural language processing model developed by Google. We fine-tuned a pre-trained BERT model on a carefully annotated dataset of student texts, enabling it to recognize linguistic patterns associated with different anxiety levels. The model achieves high accuracy in distinguishing between low, moderate, and high anxiety expressions.

The complete system follows a full-stack architecture. The trained model is deployed using a FastAPI backend, which handles prediction requests and serves responses efficiently. The frontend is built with Streamlit, providing an intuitive and accessible user interface where students can enter their thoughts and view results with visual indicators and emojis. A green smiley face indicates low anxiety, a yellow neutral face indicates moderate anxiety, and a red worried face indicates high anxiety. Along with these visual cues, the system displays personalized tips for managing anxiety based on the detected level.

We also incorporated an institutional monitoring feature that allows schools and colleges to view aggregated, anonymized anxiety trends. This helps institutions identify periods of heightened stress, such as before major examinations, and proactively organize counseling sessions, stress management workshops, or academic support programs. All data is anonymized to protect student privacy, and the system includes clear disclaimers that it is a supportive tool, not a clinical diagnostic instrument.

Throughout the development process, we placed strong emphasis on ethical considerations. Student privacy is paramount, and the system does not store any personally identifiable information. We implemented transparent communication about how the AI makes decisions and ensured that users understand the non-diagnostic nature of the tool.

➢ Achievement of Objectives

All primary objectives set at the beginning of this project have been successfully achieved. The system provides real-time anxiety detection, accurately categorizing student texts into three distinct anxiety levels. The response time is under two seconds, ensuring a smooth user experience. The user interface is intuitive and accessible from any device with a web browser, making it easy for students to use anytime and anywhere. The feedback mechanism delivers relevant and helpful coping tips tailored to each anxiety level. The institutional dashboard provides valuable insights while maintaining complete anonymity. And the entire system has been designed and built with ethical principles at its core.

➢ Key Findings and Insights

Through this project, we gained several important insights. First, we observed that BERT models are exceptionally effective at capturing subtle linguistic cues that indicate emotional states. Words, phrases, sentence structures, and even punctuation patterns all contribute to accurate anxiety detection. Students expressing low anxiety tend to use confident, positive language and focus on their preparation and readiness. Those with moderate anxiety often express some nervousness but also mention coping strategies or hope. High anxiety texts frequently contain words related to fear, panic, physical symptoms like racing heart or inability to sleep, and catastrophic thinking about failure.

Second, we found that anonymity significantly influences what students share. During our testing phases, students reported feeling more comfortable expressing their true feelings when they knew their identity would not be revealed. This confirms that privacy-preserving design is essential for mental health tools.

Third, we learned that immediate feedback has a positive impact. Students who received instant anxiety assessment along with coping tips reported feeling more aware of their emotional state and more empowered to take action. Some even shared that simply seeing their anxiety level validated made them feel less alone in their experience.

Significance of the Work

This project makes several important contributions. It is, to our knowledge, one of the first dedicated systems specifically designed for exam anxiety detection using advanced deep learning techniques. While there are general emotion detection systems and mental health chatbots available, none focus specifically on the unique context of examination stress. Our system fills this gap.

The annotated dataset we created during this project is another valuable contribution. It consists of thousands of student texts labeled with anxiety levels, which can be used for further research and development in this area.

The full-stack implementation provides a working template that can be adapted and extended by other researchers and developers interested in applying AI to mental health in educational settings.

Most importantly, this project demonstrates that AI can be used for social good. Technology is often developed for commercial purposes, but here we have shown how cutting-edge natural language processing can be harnessed to support student well-being and potentially save lives by enabling early intervention.

➢ Limitations Acknowledged

While the project has been successful, we acknowledge certain limitations. Currently, the system only supports English language input, which limits its accessibility in regions where students primarily communicate in other languages. The model's accuracy depends on the quality and diversity of the training data, and while we made efforts to create a balanced dataset, there may be cultural variations in how anxiety is expressed that are not fully captured. The system analyzes text only and does not consider other indicators of anxiety such as voice tone, facial expressions, or physiological signals. And importantly, it is a supportive tool, not a replacement for professional mental health care.

➢ Final Thoughts

The AI-Based Exam Anxiety Detector represents a meaningful step toward integrating artificial intelligence into student mental wellness support. By providing a real-time, accessible, anonymous, and scalable tool, we hope to help educational institutions identify students who need support before they reach a crisis point. Every student deserves to approach examinations with confidence and calm, and if this system can help even one student recognize their anxiety and seek help, our efforts will have been worthwhile.

The project has been a journey of learning, innovation, and deep engagement with a problem that affects millions of students worldwide. We are proud of what we have built and excited about the possibilities it opens for future work.

## FUTURE SCOPE

While the current system successfully demonstrates the core concept of AI-based exam anxiety detection, there are numerous opportunities for enhancement and expansion. The future scope of this project is vast and promising.

➢ Enhanced Language Support

One of the most important future directions is expanding language support. The current system works only with English text, but students in India and many other countries often express themselves more naturally in their regional languages. Future versions could incorporate multilingual capabilities, starting with major Indian languages such as Hindi, Tamil, Telugu, Bengali, Marathi, and others. This would require collecting annotated datasets in each language and fine-tuning multilingual BERT models or other cross-lingual transformers. Making the system available in multiple languages would dramatically increase its reach and impact.

➢ Voice Input Integration

Another valuable enhancement would be adding voice input capability. Many students might find it easier to speak their thoughts rather than type them, especially when feeling

anxious. Voice input would also capture paralinguistic features such as tone, pitch, speaking rate, and hesitations, which could provide additional signals for anxiety detection. This would involve integrating speech-to-text technology and potentially developing multimodal models that combine text and audio features for more accurate assessment.

➢ Mobile Application Development

The current web-based system works well on mobile browsers, but a dedicated mobile application could offer enhanced features and better user experience. A native mobile app could work offline, send reminders and check-ins, integrate with device sensors for additional data, and provide push notifications for timely interventions. It could also incorporate features like mood tracking over time, journaling with automatic analysis, and direct links to counseling resources.

➢ Deeper Integration with Learning Management Systems

Integrating the anxiety detector directly into existing Learning Management Systems like Moodle, Google Classroom, Canvas, or Blackboard would make it seamlessly accessible to students within their regular academic workflow. This could enable automated check-ins before major exams, with results visible to counselors (anonymized) for proactive outreach. Such integration would normalize mental health check-ins as a routine part of academic life.

➢ Advanced Analytics and Predictive Capabilities

The institutional dashboard could be enhanced with more sophisticated analytics. Machine learning models could identify patterns and predict which students or groups might be at risk based on historical data. Predictive analytics could alert institutions to upcoming stress peaks, allowing them to prepare resources in advance. Time-series analysis could track how anxiety levels change throughout the academic year and identify contributing factors.

➢ Personalized Intervention Recommendations

The feedback system could become more sophisticated by providing personalized intervention recommendations based on the student's specific anxiety profile. For example, students whose texts suggest physical symptoms might receive relaxation techniques, while those expressing catastrophic thinking might get cognitive restructuring exercises. Machine learning could even learn which interventions work best for different types of students over time.

➢ Integration with Counseling Platforms

The system could be connected directly to counseling platforms, allowing students who express high anxiety to schedule appointments with counselors immediately. For institutions with limited counseling resources, the system could help triage students, ensuring that those most in need receive priority attention.

➢ Longitudinal Tracking for Individual Students

With appropriate consent and privacy safeguards, the system could track individual students over time, building personal anxiety profiles and identifying patterns. A student might receive insights like "Your anxiety tends to peak three days before exams" or "You express more anxiety about mathematics than other subjects." This personalized insight could help students develop better self-awareness and coping strategies.

➢ Gamification and Engagement Features

To encourage regular use, the system could incorporate gamification elements. Students might earn badges for checking in regularly, complete challenges related to stress management, or unlock relaxation exercises as they engage with the platform. These features could make mental wellness practices more engaging, especially for younger students.

➢ Expanded Mental Health Focus

While the current project focuses specifically on exam anxiety, the same approach could be extended to other mental health concerns affecting students. Depression detection, burnout identification, social anxiety, and general stress monitoring could all be addressed using similar techniques. A comprehensive student mental wellness platform could emerge from this foundation.

➢ Clinical Validation and Partnership

For the system to reach its full potential, clinical validation through partnerships with mental health professionals and institutions would be valuable. Rigorous studies comparing the system's assessments with clinical evaluations would strengthen its credibility. Partnerships with schools, colleges, and mental health organizations would ensure the system meets real-world needs effectively.

➢ Continuous Model Improvement

The machine learning model itself can continuously improve as more data becomes available. Implementing a feedback loop where users can indicate whether the assessment felt accurate would provide valuable training data. Regular retraining with new data would keep the model current with evolving language patterns and expressions of anxiety.

➢ Ethical Framework Development

As the system evolves, so should its ethical framework. Future work could include developing more sophisticated privacy-preserving techniques like federated learning, where models improve based on user data without the data ever leaving users' devices. Transparent explainability features could help users understand why the system made particular assessments, building trust and understanding.

➢ Research Publications and Knowledge Sharing

Finally, sharing the findings, methodology, and lessons learned through research publications, open-source contributions, and conference presentations would benefit the broader community working at the intersection of AI and mental health. This project has the potential to inspire and inform similar efforts worldwide.

# References

1.Devlin, J., et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL.

2.Elbourhamy, D. M. (2025). Automated exam evaluation and anxiety detection. PeerJ Computer Science.

3.Hamd, M. R., et al. (2026). AI for assessing exam anxiety and academic motivation. Journal of Educational Technology.

4.Yeasmin, N., et al. (2025). AI in mental health: An umbrella review. Health Science Reports.

5.Spielberger, C. D. (1983). Manual for the State-Trait Anxiety Inventory.

6.American Test Anxieties Association. (2023). Test Anxiety Statistics.

7.NCERT. (2023). Student Mental Health Survey, India.